



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

UNIT TEST

30-05-2021

JEE(Main + Advance) : Nurture Course (PHASE : S (TEPS))

Test Type : UNIT TEST

Test Pattern : JEE (Main)

TEST DATE : 30-05-2021

ANSWER KEY

PHYSICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	C	C	D	A	C	B	B	C
SEC-A	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	B	B	A	C	C	B	B	B	B
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	15.00	130.00	500.00	412.00 to 413.00	5.00	193.00	127.00	35.00	3.50	7.00

CHEMISTRY

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	C	D	B	B	A	A	A	D
SEC-A	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	C	A	A	C	A	C	D	A	B
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	7.00	150.00	9.00	27.00	16.00	1.00	3.00	5.00	5.00	9.00

MATHEMATICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	B	D	B	A	A	A	A	B
SEC-A	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	D	B	A	A	A	D	A	A	C
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	8.00	7.00	2.00	5.00	7.00	0.00	3.00	215.00	8.00	2.00

HINT – SHEET

PHYSICS

SEC-A

1. **Ans (B)**

$$F = \frac{\text{Newton}}{\text{m}^2}$$

$$\frac{hc}{d^2} = \text{joule-sec.} \left(\frac{\text{metre}}{\text{sec.}} \right) \frac{1}{(\text{metre})^2}$$

$$= \frac{\text{joule}}{\text{metre}} = \text{Newton}$$

$$\frac{hc}{d^4} = \left(\frac{hc}{d^2} \right) \frac{1}{d^2} = \frac{\text{Newton}}{\text{m}^2}$$

Hence $F = hc/d^4$

$$\frac{hd}{c} = \frac{\text{Joule. sec} (\text{metre})^2}{\left(\frac{\text{metre}}{\text{sec.}} \right)}$$

$$= \text{Newton-metre}^2 \text{ sec}^2$$

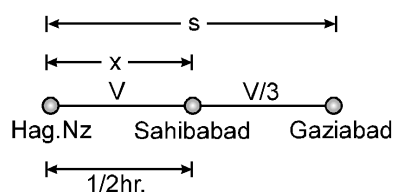
$$\frac{d^4}{hc} = \frac{1}{\left(\frac{hc}{d^4} \right)} = \frac{\text{m}^2}{\text{Newton}}$$

3. **Ans (C)**

Due to gravity, speed of particle along ABC path will increase early therefore $t_1 < t_2$.

4. **Ans (C)**

As shown in figure.



$$\frac{x}{V} = \frac{1}{2} \text{ hr.} \dots\dots\dots(1)$$

$$\frac{S}{V} = T$$

(actual time if there would not have been any accident)

$$\Rightarrow \frac{x}{V} + \frac{(S-x)}{V/3} = T + 2$$

$$\Rightarrow \frac{x}{V} - \frac{3x}{V} + \frac{3S}{V} = T + 2$$

$$\Rightarrow 3T - 1 = T + 2 \Rightarrow T = \frac{3}{2} = 1.30 \text{ min.}$$

$$\Rightarrow \text{Correct time} = S + \text{time} + 1.20 \text{ hr} = 915 + 1.30 + 2 \text{ hr} = 10.45 + 2 \text{ hr}$$

at that day it had reached 2 hrs late thus at 12.45 PM.

5. **Ans (D)**

$$\text{From graph } u_y = \frac{dy}{dt} = \tan 60^\circ$$

$$u_y = \tan 60^\circ = \sqrt{3} \text{ m/s}$$

$$\text{Range } R = \frac{2 u_x u_y}{g} \text{ or } \sqrt{3} = \frac{2 \times u_x \times \sqrt{3}}{g}$$

$$\text{or } u_x = 5 \text{ m/s}$$

$$\therefore u = \sqrt{u_x^2 + u_y^2} = \sqrt{28} \text{ m/s}$$

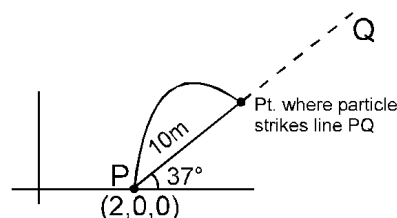
6. **Ans (A)**

$$\text{Range} = 10 \text{ m.}$$

For point where particle strikes line PQ

$$\therefore x \text{ coordinate} = 10 \cos 37^\circ + 2 = 10 \text{ m}$$

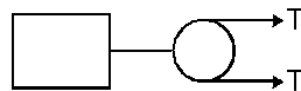
$$y \text{ coordinate} = 10 \sin 37^\circ = 6 \text{ m}$$



$$z \text{ coordinate} = 0 \text{ m}$$

7. **Ans (C)**

F.B.D of the mass M :



$$T = F$$

$$ma = 2T = 2F$$

$$\Rightarrow a = \frac{2F}{m}$$

so, acceleration of point P is $2a = \frac{4F}{m}$ (From constraint relation)

8. **Ans (B)**

$$I_1 = \frac{m\ell^2}{12}$$

$$I_2 = m \left(\frac{\ell}{2\pi} \right)^2 = \frac{m\ell^2}{4\pi^2}$$

$$\frac{I_1}{I_2} = \frac{m\ell^2}{12} : \frac{m\ell^2}{4\pi^2} = \pi^2 : 3$$

9. **Ans (B)**

$$W = \Delta K$$

$$= 0 - \frac{1}{2} 2mR^2 \left(\frac{v}{R} \right)^2$$

$$W = -mv^2$$

$$W_{\text{ext}} = mv^2$$

10. **Ans (C)**

Suppose m gram of ice converted into water.

$$\Delta V = \frac{m}{917} - \frac{m}{1000} = 0.125 \times 10^{-6}$$

$$m = \frac{125 \times 917}{83} \times 10^{-6}$$

Heat lost by metal = Heat absorbed by ice

$$10^{-2} \times 5 \times 100 = m \times 336 \times 10^3$$

$$S = \frac{125 \times 917 \times 336 \times 10^{-3}}{83}$$

$$S = 464$$

12. **Ans (B)**

Let w be the water equivalent.

The rate of cooling in the first case is

$$\frac{(50 + w)(50 - 45)}{10}$$

In the second case, rate of cooling

$$= \frac{(100 + w)(50 - 45)}{18}$$

Since they cool under identical conditions

$$(50 + w) \frac{5}{10} = (100 + w) \frac{5}{18}$$

$$w = 12.5 \text{ gms}$$

13. **Ans (B)**

When the temperature decreases the wire tends to contract in length and lift the mass up. But the weight begins to produce a stress and counteracts the contraction produced.

Let $\Delta \ell$ be the stretching produced by weight W .

$$\Delta L =$$

$$\frac{WL}{AY} = \frac{2 \times 9.8 \times 3}{\pi(0.25)^2 \times 10^{-6} \times 206 \times 10^9}$$

Contraction due to cooling = $L\alpha\theta$

$$= 3 \times 11 \times 10^{-6} \times \theta$$

$$3 \times 11 \times 10^{-6} \theta =$$

$$\frac{2 \times 9.8 \times 3}{\pi(0.25)^2 \times 10^{-6} \times 206 \times 10^9}$$

$$= \frac{2 \times 9.8 \times 10^3}{\pi(0.25)^2 \times 206 \times 11}$$

$$= 44^\circ\text{C}$$

14. **Ans (A)**

Internal energy of n moles of a perfect gas at temperature T is given by

$$U = nC_v T$$

$$C_p - C_v = \frac{R}{J} \text{ and } \frac{C_p}{C_v} = \gamma$$

$$C_v = \frac{R}{\gamma - 1}$$

$$U = \frac{nRT}{\gamma - 1}$$

$$pV = nRT$$

$$U = \frac{pV}{\gamma - 1} = \frac{10^5 \times 100}{1.4 - 1} = \frac{10^7}{0.4}$$

$$= 2.5 \times 10^7 \text{ J}$$

16. **Ans (C)**

Rate of loss of energy by unit area of the planet

$$= \sigma T^4 \text{ where } \sigma \text{ is}$$

Stefan's constant.

Let Q be the total energy emitted by the Sun in every second. If d is the distance of the planet from Sun, then Q falls uniformly over the inner surface of the sphere of radius d .

Rate of gain of heat by unit area of planet

$$= \frac{Q}{4\pi d^2}$$

For steady temperature of planet,

$$\sigma T^4 = \frac{Q}{4\pi d^2}$$

$$T^4 = \frac{Q}{4\pi\sigma d^2} \text{ or } T = \left(\frac{Q}{4\pi\sigma d^2} \right)^{1/4}$$

$$T \propto \frac{1}{\sqrt{d}}$$

17. **Ans (B)**

During the adiabatic process,

$$p^\gamma - T^\gamma = \text{constant}$$

$$\frac{T^\gamma}{p^{\gamma-1}} = \text{constant}$$

$$p\gamma^{-1} = T\gamma/\text{constant}$$

$$p \propto T^{\gamma/\gamma-1}$$

$$\text{Hence } p T^3 \therefore \frac{\gamma}{\gamma-1} = 3 \Rightarrow \gamma = \frac{3}{2}$$

18. **Ans (B)**

Consider an element of rod at a distance x and thickness dx .

$$\text{Then, } Q = KA \cdot \frac{dT}{dx} = \frac{CA}{T} \cdot \frac{dT}{dx}$$

$$q \text{ (heat flow density)} = \frac{Q}{A} = \frac{C}{T} \cdot \frac{dT}{dx}$$

$$\text{Integrating, } q \int_0^{\ell} dx = C \int_{T_1}^{T_2} \frac{dT}{T}$$

$$q\ell = C \log \frac{T_2}{T_1} \text{ or } \frac{C}{\ell} \log \left(\frac{T_2}{T_1} \right)$$

19. **Ans (B)**

AB $\rightarrow$ constant P increasing V

$\therefore$ increasing T

BC $\rightarrow$ constant T increasing V

$\therefore$ decreasing P

CD $\rightarrow$ constant V decreasing P

$\therefore$ decreasing T

DA $\rightarrow$ constant T decreasing V

$\therefore$ increasing P

Also BC is at a higher temperature than AD.

20. **Ans (B)**

$$W_{AB} = W_{CD} = 0$$

$$W_{\text{total}} = P_B(V_L - V_B) - P_A(V_D - V_A)$$

$$= nR(T_C - T_B) - nR(T_D - T_A)$$

$$= nR 1200 - nR 600$$

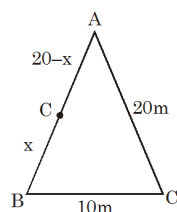
$$= nR 600$$

$$= 3 \times \frac{25}{3} \times 600 = 15000\text{J} = 15\text{kJ}$$

PHYSICS

SEC-B

1. **Ans (15.00)**



As c moves with speed 5 m/s towards 'B' at any instant, therefore distance covered by 'C' in ' t ' sec

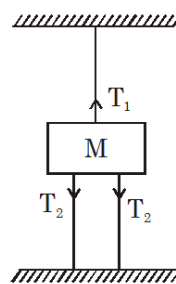
$$d_{BC} \quad 10 - x = 5t$$

$$20 - (20 - x) = 5t$$

$$t = 1 \text{ sec}$$

$$d_{AC} = 15 \text{ m}$$

2. **Ans (130.00)**



$$T_1 = Mg + 2T_2$$

$$= 100 + 10 = 110 \text{ N}$$

When elevator is moving upward with 3 m/s^2

$$T_1 + x = M(g + a) + (T_2 - x) \times 2$$

$$110 + x = 130 + 10 - 2x$$

$$3x = 30 \Rightarrow x = 10$$

Lower string will slack

Tension in upper string = 130 N

3. **Ans (500.00)**

Since no external torque acts on the system,

$$I_1\omega_1 = I_2\omega_2$$

$$\left[\frac{100 \times (1.5)^2}{2} + 50 \times (1.5)^2 \right] \omega$$

$$= \frac{100 \times (1.5)^2}{2} \omega$$

$$\Rightarrow \omega = \frac{\omega}{2}$$

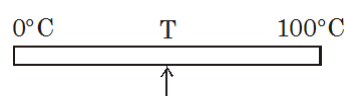
$$\Delta k = \frac{1}{2} \times 50 \times \frac{9}{4} \times 4\omega^2$$

$$- \frac{1}{2} \times 100 \times \frac{9}{4} \omega^2$$

$$\frac{225}{2} \times \left(\frac{20 \times 2\pi}{60} \right)^2$$

$$\frac{25}{2} \times 4\pi^2 = 500$$

4. **Ans (412.00 to 413.00)**



$$25 = \frac{400 \times 10^{-4}}{0.5} [(T - 0) + (T - 100)]$$

$$50 + \frac{2500}{16} = T$$

$$\frac{dT}{dx} = \frac{T - 0}{0.5} = 412.5$$

5. **Ans (5.00)**

$$\frac{3}{2} kT = 6 \times 10^{-21}$$

$$PV = \frac{N}{N_a} N_a kT$$

$$2 \times 10^5 \times 10^{-4} = N 4 \times 10^{-21}$$

$$N = 5 \times 10^{21}$$

6. **Ans (193.00)**

Volume of bulb = $V \text{ mm}^3$

Total volume = $V + 3$

$$(V + 3) (1 + 24 \times 10^{-6} \times 100)$$

$$= V (1 + 1.8 \times 10^{-4} \times 100)$$

$$V + 3 + (V + 3) (24 \times 10^{-4})$$

$$= V + V (180 \times 10^{-4})$$

$$3 + 72 \times 10^{-4} = V 156 \times 10^{-4}$$

$$V = \frac{1000}{52} + \frac{6}{13} = 192.8 \text{ mm}^3$$

7. **Ans (127.00)**

Suppose 'x' unit converted into ice

$$1 \times 4.2 \text{ K} \times 10 - x 333 \times 10^3 = 0$$

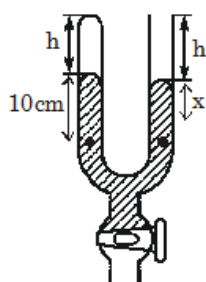
$$x = \frac{4.2 \times 1000}{330} \text{ gram}$$

$$= \frac{140}{11} = 127.00$$

8. **Ans (35.00)**

Initial pressure in left side = p_0

& final = p_1



$$p_0 A h = p_1 A (h + 10)$$

$$p_1 = p_0 + \rho g (10 - x)$$

$$p_0 h = [p_0 + \rho g (10 - x)] (h + 10)$$

$$0 = 10^4 + 13.6 \times 10^4 (0.1 - x) (0.3)$$

$$x = \frac{10^4 (1 + 0.408)}{10^4 (40.8)}$$

$$= \frac{14.08}{40.8} = \frac{1408}{40.8} \text{ cm}$$

$$x = 35 \text{ cm}$$

9. **Ans (3.50)**

$$T_f = \frac{0.1 \times 87 + 0.2 \times 27}{0.3} = 47^\circ \text{C}$$

$$\Delta S = \int \frac{dQ}{T} = \int \frac{msdT}{T} = ms \ln \frac{T_f}{T_i}$$

$$\Delta S = -0.1 \times s \ln \left(\frac{360}{320} \right) + 0.2 s \ln \left(\frac{320}{300} \right) = 3.5$$

10. **Ans (7.00)**

$$Q_{in} = nC_V (T_A - T_C) + nRT_A \ln \left(\frac{1.8}{1.2} \right)$$

$$= \frac{3R}{2} \left(\frac{340}{3} \right) + R 340 \times 0.4$$

$$= 340 \times 0.9 R$$

$$Q_{out} = \frac{-n5R}{2} \frac{340}{3}$$

$$\eta = \left(1 - \frac{5/6}{0.9} \right) \times 100 = \left(1 - \frac{50}{54} \right) \times 100$$

$$= \frac{400}{54} = 7.4$$

CHEMISTRY

SEC-A

1. **Ans (B)**

Total Energy in 1 min = $5 \times 60 = 300 \text{ J}$

$$\text{Total Energy} = n \times \frac{hc}{\lambda}$$

$$n = \frac{300 \times \lambda}{hc} = \frac{300 \times 198 \times 10^{-9}}{6.62 \times 10^{-34} \times 3 \times 10^8}$$

$$n = 3 \times 10^{20}$$

3. **Ans (C)**

Boyle's law is also called constant temperature law.

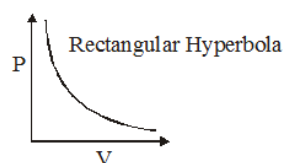
For an ideal gas at an particular temperature the pressure becomes inversally proportional to volume of gas.

$$PV = nRT \text{ (Ideal gas equation)}$$

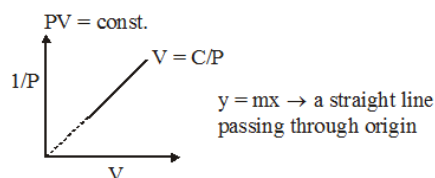
If temperature is kept constant.

$$PV = \text{constant}$$

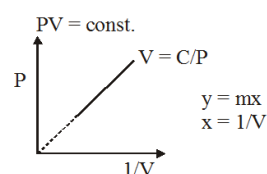
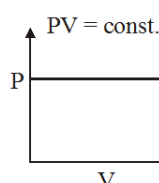
$$P \propto \frac{1}{V}$$



PV v/s curve



Pv/s 1/V curve



Option (C) does not represent the boyles law.

4. **Ans (D)**

$$P_{\text{gas}} = 1 + 0.5 = 1.5 \text{ atm}$$

$$\log(nRT) = \log 3$$

$$= nRT = 3$$

$$\Rightarrow V = \frac{3}{1.5}$$

5. **Ans (B)**

$$n = 5$$

$$5 \rightarrow 1, 4 \rightarrow 1, 3 \rightarrow 1, 2 \rightarrow 1$$

6. **Ans (B)**

Let normally of KMnO_4^- solution is N

$$\therefore N \times 10 = 50 \times 1 \Rightarrow N = 5$$

molecular equation of MnO_4^- = molecular equation of oxalic acid

$$5 \times 100 = \frac{W}{E} \times 1000 \Rightarrow \left(\frac{W}{90/2} \right) \times 1000 W$$

$$= 22.5 \text{ gm Ans.}$$

7. **Ans (A)**

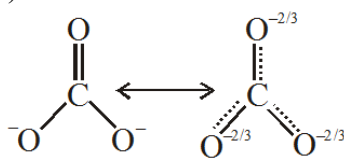
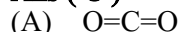
Normality of $\text{NaOH} = x$

Molarity = x

$$N_{\text{Ba(OH)}_2} = Y, M = \frac{Y}{2}$$

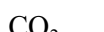
$$V_1 \times X + V_2 \times Y = 100 \times 0.1$$

12. **Ans (C)**

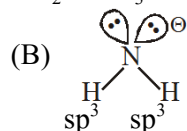


Bond angle = 180°

$\text{CO}_2 > \text{CO}_3^{2-}$



120°



$< \text{NH}_3$

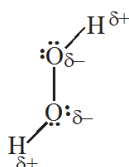
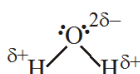
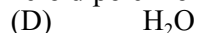
Bond angle

more lone pair on NH_2^- lead to more repulsion and decrease in bond angle.

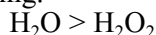


Correct

Since Sn has lone pair bent shape so non zero dipole moment.

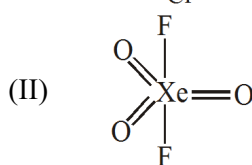
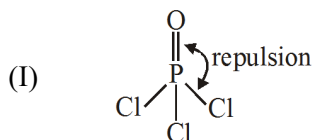


H_2O has more negative charge accumulated on oxygen so more strength of H-Bonding.

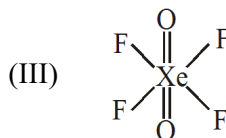


Option (C) is correct.

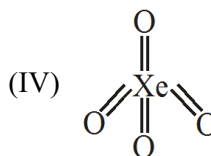
13. **Ans (A)**



Symmetrical structure so net repulsion equal zero.



Symmetrical structure so net repulsion equal zero.



Equivalent hybridization

Double bond will not affect structure in case of II, III, IV

Option (A) is correct.

14. **Ans (A)**

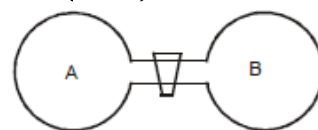
$$\text{L.E.} + 2P + 3Q = \Delta H_{\text{sol.}}$$

$$P + \frac{3}{2} Q = \frac{\Delta H_{\text{sol.}} - \text{L.E.}}{2}$$

CHEMISTRY

SEC-B

1. **Ans (7.00)**



Pressure in A Bulb after mixing two gases is ' P_2 '

$$P_1 V_1 = P_2 V_2$$

$$9 \times 5 = 15 \times (P_2)$$

$$3 \text{ atm} = P_2$$

Pressure in B Bulb after mixing two gases is ' P_2 '

$$P_1 V_1 = P_2 V_2$$

$$6 \times 10 = (P_2) \times 15$$

$$4 \text{ atm} = P_2$$

$$\text{Total Pressure of Bulb} = 4 + 3 = 7 \text{ atm.}$$

2. Ans (150.00)

$$\text{Given : } X_{\text{CaBr}_2} = \frac{n_{\text{CaBr}_2}}{n_{\text{CaBr}_2} + n_{\text{NaI}}}$$

Moles fraction :

$$\therefore \frac{1}{3} = \frac{n_{\text{CaBr}_2}}{n_{\text{CaBr}_2} + n_{\text{NaI}}} \left\{ \begin{array}{l} \text{NaI \& CaBr}_2 \\ \text{Solute} \end{array} \right\}$$

$$\therefore \boxed{n_{\text{CaBr}_2} = \frac{n_{\text{NaI}}}{2}}$$

⇒ If assume ; moles of NaI = x moles

$$\therefore \text{moles of CaBr}_2 = \frac{x}{2} \text{ moles}$$

$$\therefore \text{Molar mass of NaI} = 23 + 127 = 150 \text{ g/mol}$$

$$\text{Molar mass of CaBr}_2 = (80 \times 2) + 40 = 200$$

g/mol

$$\Rightarrow \text{Volume of solution} = 1 \text{ L} = 1000 \text{ mL}$$

$$\therefore \text{Mass of solution} = \text{Volume of solution} \times$$

density of solution

$$= 1000 \times 1.15$$

$$= 1150 \text{ g}$$

$$\therefore \text{Mass of solvent} = \text{Mass of solution} - \text{Mass}$$

of solute

$$= 1150 - (x \times 150 + 200 \times \frac{x}{2})$$

$$= (1150 - 250x) \text{ gm}$$

$$\therefore m = \frac{\text{moles of NaI}}{\text{mass of solvent (gm)}} \times 1000$$

$$\Rightarrow \frac{10}{9} = \frac{x}{(1150 - 250x)} \times 1000$$

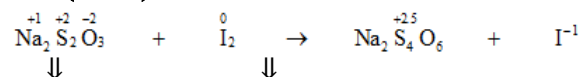
$$\Rightarrow \text{On solving } \boxed{x = 1}$$

$$\therefore \text{Mass of NaI} = \text{Moles} \times \text{molar mass}$$

$$= 1 \times 150$$

$$= 150 \text{ gm}$$

3. Ans (9.00)



$$(n_f = 1) \quad \quad \quad (n_f = 2)$$

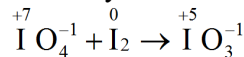
$$\therefore \text{Meq. of Na}_2\text{S}_2\text{O}_3 = \text{meq. of I}_2$$

$$\Rightarrow M \times V_{\text{mL}} \times n_f = \text{mmoles} \times n_f$$

$$\therefore \text{m moles of I}_2 = \frac{0.1 \times 10000 \times 1}{2}$$

$$\therefore \text{m moles of I}_2 = 50 \rightarrow \text{from 1}^{\text{st}} \text{ reaction}$$

Similarly for 2nd reaction



$$n_f = 2 \quad n_f = 10$$

$$\Rightarrow \text{meq. of IO}_4^{-1} = \text{meq. of I}_2$$

$$M \times V_{\text{mL}} \times n_f = (\text{m moles}) \times n_f$$

$$\therefore \text{m moles of I}_2 = \frac{0.2 \times 1000 \times 2}{10}$$

$$\therefore \text{m moles of I}_2 = 40 \text{ from 2}^{\text{nd}} \text{ reaction}$$

$$\text{total m moles of I}_2 = 50 + 40 = 90 \text{ m moles}$$

$$\text{OMR Ans.} = 9 + 0 = 9$$

5. Ans (16.00)

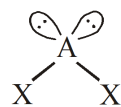
(i) AX_{n1} A belongs to the group 16th oxygen

finally planer and Polar

AX₂ Such as H₂O or OF₂ Like compounds

$$n_1 = 2$$

$$m_1 = 2 \rightarrow \text{No. of lone pairs}$$

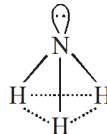


(ii) BX_{n2} Characteristics

* Group 15th Nitrogen family

* Trigonal pyramidal

BX₃ Like NH₃ or PCl₃



$$n_2 = 3$$

$$m_2 = 1 \text{ no. of lone pair}$$

(iii) CX_{n3} Characteristics

* Belongs to group 14th Carbon family

* Zero dipole moment

CH₄, CCl₄ etc.

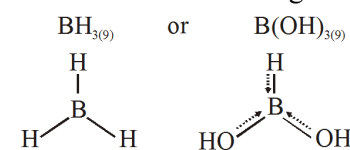
$$n_3 = 4, m_3 = 0$$

(iv) DX_{n4} Characteristics

* Central atom belongs to 13th group

Boron family.

* All bond angles are 120°



$$n_4 = 3, m_4 = 0$$

$$\left| \frac{n_1 + n_2 + n_3 + n_4}{m_1 + m_2 + m_3 + m_4} \right| = \left| \frac{2+3+4+3}{2+1+0+0} \right| = |4|^2 = 16 \text{ Answer}$$

6. Ans (1.00)

(A) $Mg = 1s^2 2s^2 2p^6 3s^2$; $Al = 1s^2 2s^2 2p^6 3s^2 3p^1$ As electron is to be removed from stable completely filled s-orbital of Mg as compared to partially filled p-orbital of Al.

(B) Li^+ due to small in size attracts more no. of water molecules and thus have bigger hydrated ion.

(C) Addition of $2^{nd} e^-$ to an anion (same charge) is difficult due to the electrostatic repulsion.

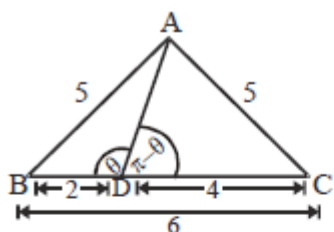
(D), (F) and (G) are correct statements

(E) First decreases from B to Al and then increases marginally owing to discrepancies in atomic size of the element (INCORRECT).

MATHEMATICS

SEC-A

1. Ans (D)



Let $AD = l$ (from figure)

$$\cos \theta = \frac{4 + l^2 - 25}{4l} \quad \&$$

$$\cos (\pi - \theta) = \frac{16 + l^2 - 25}{8l}$$

$$\Rightarrow l^2 - 21 + \frac{l^2 - 9}{2} = 0 \Rightarrow l = \sqrt{17}$$

2. Ans (B)

$$\sin^2 x - c \sin x + \sin x + 2c - 6 \geq 0$$

$$(\sin x - 2)(\sin x + 3 - c) \geq 0$$

$$\sin x + 3 - c \leq 0$$

$$c - 3 \geq \sin x$$

$$c \geq 4$$

3. Ans (B)

$$y^2 + 2y + 1 + \sec^2 \left(\frac{\pi}{2}(y+x) \right) - 1 = 0$$

$$(y+1)^2 + \tan^2 \left(\frac{\pi}{2}(y+x) \right) = 0$$

$$y = -1 \text{ and } \frac{\pi}{2}(x-1) = n\pi$$

$$x-1 = 2n, \quad x = 2n+1, \quad \text{odd}$$

4. Ans (D)

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = b - c + c - a + a - b = 0$$

$$\left(\frac{x}{a} \right)^3 + \left(\frac{y}{b} \right)^3 + \left(\frac{z}{c} \right)^3 = 3 \left(\frac{x}{a} \right) \left(\frac{y}{b} \right) \left(\frac{z}{c} \right)$$

5. Ans (B)

$$y + \sqrt{3}|x| = 2$$

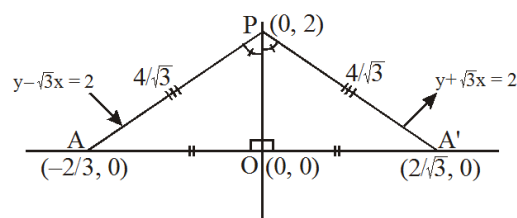
$$\Rightarrow y + \sqrt{3}x = 2 \quad \text{if } x \geq 0$$

$$\& \quad y - \sqrt{3}x = 2 \quad \text{if } x < 0$$

$$\text{if } x \geq 0$$

$$y + \sqrt{3}x = 2$$

$$\Rightarrow \frac{y}{2} + \frac{x}{2/\sqrt{3}} = 1$$



$$\text{if } x < 0$$

$$y - \sqrt{3}x = 2$$

$$\Rightarrow \frac{y}{2} - \frac{x}{2/\sqrt{3}} = 1$$

$$PA = \sqrt{(2)^2 + \left(\frac{2}{\sqrt{3}} \right)^2} = \sqrt{\frac{16}{3}} = \frac{4}{\sqrt{3}}$$

$$|AO| = |A'O|$$

Hence, $\angle APO = \angle A'PO$

Since PO is the angle bisector and AO is perpendicular to PO, so O is the intersection point having co-ordinates (0, 0).

$\Rightarrow$ Option (B) is correct.

6. **Ans (A)**

$$x^2 + y^2 = 4$$

let midpoint of any chord of circle (h, k)

$$\text{equation of chord is } hx + ky = h^2 + k^2$$

using homogenisation

$$x^2 - 2(x + y) \left(\frac{hx + ky}{h^2 + k^2} \right) = 0$$

$$(h^2 + k^2)x^2 - 2(hx^2 + kxy + hxy + ky^2) = 0$$

$$(h^2 + k^2 - 2h)x^2 - 2ky^2 - 2(h + k)xy = 0$$

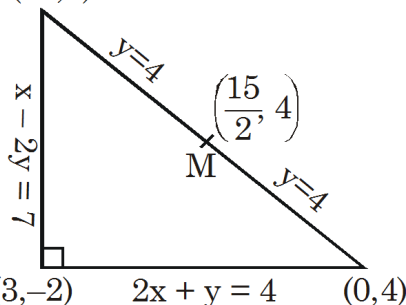
lines are at right angle, so

$$(h^2 + k^2 - 2h) + (-2k) = 0$$

$$h^2 + k^2 - 2h - 2k = 0$$

7. **Ans (A)**

$(15, 4)$

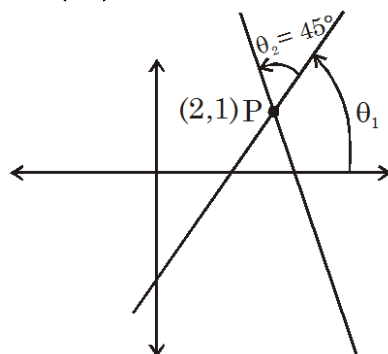


orthocentre : $(3, -2)$

circumcentre : $\left(\frac{15}{2}, 4\right)$

$$\text{Slope} = \frac{4 + 2}{\frac{15}{2} - 3} = \frac{12}{9} = \frac{4}{3}$$

8. **Ans (A)**



$$\therefore \tan \theta_1 = 2$$

let slope of new line is m

$$\therefore m = \tan(\theta_1 + \theta_2)$$

$$\Rightarrow m = \frac{\tan \theta_1 + \tan \theta_2}{1 - \tan \theta_1 \tan \theta_2}$$

$$\Rightarrow m = \frac{2 + 1}{1 - 2 \cdot 1} = -3$$

$$\therefore \text{equation} \Rightarrow y - 1 = -3(x - 2)$$

$$\Rightarrow 3x + y = 7$$

9. **Ans (A)**

It is a radical centre

$$S_1 - S_2 = 0 \Rightarrow -5x + 11y - 7 = 0 \quad \dots(1)$$

$$S_1 - S_3 = 0 \Rightarrow 3x - 5y + 1 = 0 \quad \dots(2)$$

from (1) and (2) $x = 3, y = 2$

10. **Ans (B)**

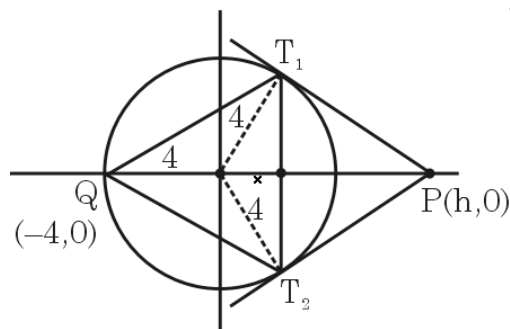
$$3\ell^2 + 6\ell + 1 = 6m^2$$

$$(3\ell + 0m + 1)^2 = 6(m^2 + \ell^2)$$

$$\frac{|3\ell + 0m + 1|}{\sqrt{m^2 + \ell^2}} = \sqrt{6}$$

$$\Rightarrow \text{required circle is } (x - 3)^2 + y^2 = 6$$

11. **Ans (A)**



$$x^2 + y^2 = 16$$

equation of T_1T_2

$$hx = 16 \Rightarrow x = \frac{16}{h}$$

$$\frac{-4 + h}{2} = \frac{16}{h} \Rightarrow -4h + h^2 = 32$$

$$\Rightarrow h^2 - 4h - 32 = 0$$

$$\Rightarrow h = 8, -4 \text{ (reject)}$$

$$x^2 + y^2 = 16, x = 2$$

$$y^2 = 12 \Rightarrow y = \pm 2\sqrt{3}$$

$$T_1 (2, 2\sqrt{3})$$

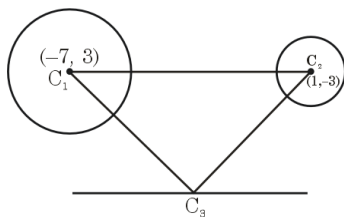
$$T_2 (2, -2\sqrt{3})$$

$$\text{Area} = \frac{1}{2} d_1 d_2$$

$$= \frac{1}{2} (12) (4\sqrt{3}) = 24\sqrt{3}$$

12. Ans (D)

$$C_1 \equiv (-7, 3), C_2 \equiv (1, -3)$$



$C_1C_2 + C_2C_3 + C_3C_1$ will be minimum if $C_1C_3 = C_2C_3$

$\Rightarrow C_3$ is point of intersection of $3x + 4y - 16 = 0$

and perpendicular bisector of $(-7, 3), (1, -3)$

$$\text{i.e. } 4x - 3y + 12 = 0$$

$$\therefore C_3 \equiv (0, 4)$$

Also radius of required circle

$$= \sqrt{49 + 1} - \sqrt{18}$$

$$= 5\sqrt{2} - 3\sqrt{2}$$

$$= 2\sqrt{2}$$

$\therefore$ Required circle is

$$(x-0)^2 + (y-4)^2 = 8$$

$$\Rightarrow x^2 + y^2 - 8y + 8 = 0$$

$$\therefore a=0, b=-8, c=8$$

$$\therefore a+b+c=0$$

13. Ans (B)

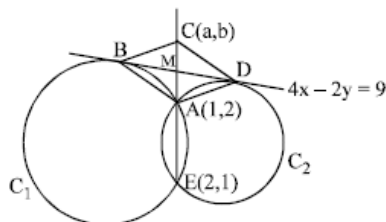
The radical axis bisects the common tangent BD.

Hence M is the mid point of BD

let $C(a, b)$

Now (a, b) lies on common chord AE which is

$$y-2 = -1(x-1)$$



$$\Rightarrow y-2 = -x+1$$

$$\Rightarrow x+y=3$$

$$\therefore a+b=3 \quad \dots(1)$$

Also $M\left(\frac{a+1}{2}, \frac{b+2}{2}\right)$ lies on $4x-2y=9$

$$4\left(\frac{a+1}{2}\right) - 2\left(\frac{b+2}{2}\right) = 9$$

$$\Rightarrow 2a+2-b-2=9 \Rightarrow 2a-b=9 \quad \dots(2)$$

Solving (1) and (2)

$$a=4 \text{ and } b=-1 \Rightarrow C(a, b) \equiv C(4, -1)$$

14. Ans (A)

$$x^3 + 2x^2 + 5x + m = 0$$

$$x^3 + 2x^2 + 3x + n = 0$$

$$x = \frac{n-m}{2}$$

and product of un common roots

$$\frac{mn}{\left(\frac{n-m}{2}\right)^2} = -1$$

$$\Rightarrow m+n=0$$

15. Ans (A)

$$195 \{(3x^3)^3 - 1\} + 3(1 + 3x^3 + 9x^6) = 0$$

$$195 \{(3x^3)^3 - 1\} + 3 \left\{ \frac{1 - (3x^3)^3}{1 - 3x^3} \right\} = 0$$

$$195 \{(3x^3)^3 - 1\} \{3x^3 - 1\} + 3 \{(3x^3)^3 - 1\} = 0$$

$$((3x^3)^3 - 1) \{195(3x^3 - 1) + 3\} = 0$$

No natural solution.

16. Ans (A)

$$\sqrt{x^2 + \sqrt{x^3 + 1}} = 1 - x$$

$$\Rightarrow x^2 + \sqrt{x^3 + 1} = 1 + x^2 - 2x$$

$$\Rightarrow \sqrt{x^3 + 1} = 1 - 2x$$

$$\Rightarrow x^3 + 1 = 1 + 4x^2 - 4x$$

$$\Rightarrow x^3 - 4x^2 + 4x = 0$$

$$\Rightarrow x(x^2 - 4x + 4) = 0$$

$$\Rightarrow x(x-2)^2 = 0$$

$$\Rightarrow x = 0, 2, 2$$

But $x=2$ is rejected

$\therefore x=0$ is only solution.

17. Ans (D)

x, A_1, A_2, y in A.P.

$$A_1 - x = y - A_2$$

$$\Rightarrow A_1 + A_2 = x + y$$

x, G_1, G_2, y in G.P.

$$\frac{G_1}{x} = \frac{y}{G_2} \Rightarrow G_1 G_2 = xy$$

x, H_1, H_2, y in H.P.

$$\frac{1}{x}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{y} \text{ in A.P.}$$

$$\frac{1}{H_1} - \frac{1}{x} = \frac{1}{y} - \frac{1}{H_2}$$

$$\frac{1}{H_1} + \frac{1}{H_2} = \frac{1}{x} + \frac{1}{y}$$

$$\frac{x+y}{xy} = \frac{A_1+A_2}{G_1 G_2}$$

$$\frac{1}{H_1} + \frac{1}{H_2} = \frac{A_1+A_2}{G_1 G_2}$$

18. Ans (A)

General term $t_n =$

$$\frac{1^2 + 2^2 + \dots + n^2}{n(n+1)} = \frac{2n+1}{6}$$

$$\therefore S_p = \sum_{n=1}^p \frac{2n+1}{6} = \frac{(p+1)^2 - 1}{6}$$

$$\text{Given } S_p + S_q - S_r = -\frac{1}{6}$$

$$\Rightarrow (p+1)^2 + (q+1)^2 - (r+1)^2 = 0$$

$$\text{or } (p+1)^2 + (q+1)^2 = (r+1)^2$$

$\Rightarrow p+1, q+1, r+1$ forms a pythagorean

triplet. The smallest values for $p+1, q+1, r+1$

are 3, 4 & 5.

$$\therefore \text{Smallest value of } p+q+r = 9$$

19. Ans (A)

$$S = 1 + 2 \left(1 + \frac{1}{50}\right) + 3 \left(1 + \frac{1}{50}\right)^2 + \dots$$

$$+ 50 \left(1 + \frac{1}{50}\right)^{49}$$

$$\left(1 + \frac{1}{50}\right) S = \left(1 + \frac{1}{50}\right) + 2 \left(1 + \frac{1}{50}\right)^2 + \dots$$

$$+ 49 \left(1 + \frac{1}{50}\right)^{49} + 50 \left(1 + \frac{1}{50}\right)^{50}$$

$$\frac{-S}{50} = 1 + \underbrace{\left(1 + \frac{1}{50}\right) + \left(1 + \frac{1}{50}\right)^2 + \dots + \left(1 + \frac{1}{50}\right)^{49}}_{50 \text{ terms}}$$

$$- 50 \left(1 + \frac{1}{50}\right)^{50}$$

$$\frac{-S}{50} = \frac{\left(\left(1 + \frac{1}{50}\right)^{50} - 1\right)}{\frac{1}{50}} - 50 \left(1 + \frac{1}{50}\right)^{50}$$

$$\frac{-S}{50} = 50 \left(1 + \frac{1}{50}\right)^{50} - 50 - 50 \left(1 + \frac{1}{50}\right)^{50}$$

$$S = 50 \times 50$$

$$S = 2500$$

Ans. (A)

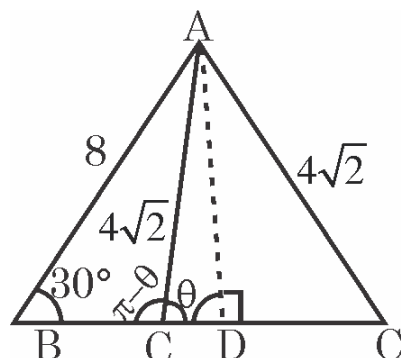
20. Ans (C)

$$\frac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C} = 8 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

MATHEMATICS

SEC-B

1. Ans (8.00)



$$\text{Area of } \triangle ABC = \frac{1}{2} \times AD \times BC$$

$$\text{Area of } \triangle ABC' = \frac{1}{2} \times AD \times BC'$$

$$\text{Difference} = \frac{1}{2} \times AD (BC' - BC)$$

$$\frac{8}{\sin \theta} = \frac{4\sqrt{2}}{\sin 30^\circ}$$

$$\sin \theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$$

$$AD = AC \sin \theta = 4$$

$$CD = AC \cos \theta = 4$$

$$\text{Difference} = \frac{1}{2} \times 4 \times 8 = 16$$

2. Ans (7.00)

$$(\operatorname{cosec}^2 \theta)^3 + (\sin^2 \theta)^3 + m^3 = 3 \operatorname{cosec}^2 \theta \cdot \sin^2 \theta \cdot m$$

$$\Rightarrow \text{either } \operatorname{cosec}^2 \theta = \sin^2 \theta = m = 1 \dots (1)$$

$$\text{or } \operatorname{cosec}^2 \theta + \sin^2 \theta + m = 0 \Rightarrow m \in (-\infty, -2]$$

$$\text{also } \frac{m(m-2)(m+1)}{(m+8)(m-7)} \geq 0$$

$$\Rightarrow m \in (-8, -1] \cup [0, 2] \cup (7, \infty)$$

from (1), (2) & (3)

integral values of m are

$$= -7, -6, -5, -4, -3, -2, 1$$

3. Ans (2.00)

$$1 - 8(\log_{10} x)^2 = \log_{10} x - 2(\log_{10} x)^2$$

$$\Rightarrow 6(\log_{10} x)^2 + \log_{10} x - 1 = 0$$

$$\Rightarrow \log_{10} x = \frac{1}{3} \text{ and } \log_{10} x = -\frac{1}{2}$$

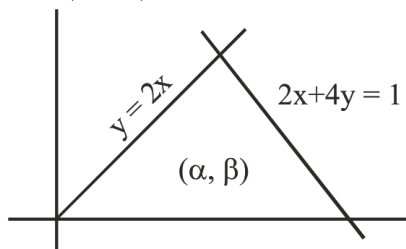
$$\therefore x = 10^{\frac{1}{3}}, x = 10^{-\frac{1}{2}}$$

$$\Rightarrow 10^{-\frac{1}{2}} < 10^{\frac{1}{3}}$$

$$\therefore \alpha = 10^{-\frac{1}{2}}, \beta = 10^{\frac{1}{3}} \quad \therefore$$

$$\alpha^2 = \frac{1}{10}, \beta^3 = 10$$

4. Ans (5.00)



$$\Rightarrow \beta + \frac{2\alpha - \beta}{\sqrt{5}} + \frac{1 - 2\alpha - 4\beta}{2\sqrt{5}} = 1$$

$$\Rightarrow 2\sqrt{5}\beta + 4\alpha - 2\beta + 1 - 2\alpha - 4\beta = 2\sqrt{5}$$

$$\Rightarrow 2\alpha + (2\sqrt{5} - 6)\beta = 2\sqrt{5} - 1$$

$$\therefore b - a = 2\sqrt{5} - 1 - 2\sqrt{5} + 6 = 5$$

5. Ans (7.00)

Members of POSL will be making angles 45°

$$\text{with } 2x + 3y - 6 = 0$$

$$\Rightarrow \tan 45^\circ = \left| \frac{m + 2/3}{1 - 2m/3} \right| = 1$$

$$\Rightarrow |3m + 2|^2 = |3 - 2m|^2$$

$$\Rightarrow 9m^2 + 12m + 4 = 9 - 12m + 4m^2$$

$$\Rightarrow 5m^2 + 24m - 5 = 0$$

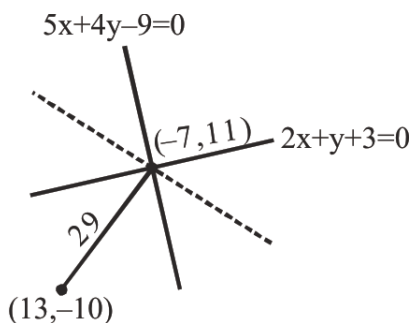
$$\Rightarrow 5\left(\frac{y}{x}\right)^2 + 24\left(\frac{y}{x}\right) - 5 = 0$$

$$\therefore \text{lines are } 5x^2 - 24xy - 5y^2 = 0$$

$$\text{Clearly } h = 12 \text{ and } a = 5$$

$$\therefore h - a = 7$$

6. Ans (0.00)



Intersection point of

$$2x + y + 3 = 0$$

$$\& 5x + 4y - 9 = 0$$

is $(-7, 11)$

hence maximum distance of point $(13, -10)$

& any member of family is '29' but the given

distance

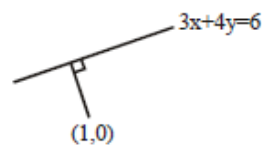
$$\left(28 + \alpha + \frac{1}{\alpha}\right), (\alpha \in \mathbb{R}^+)$$

is more than or equal to 30 because $\alpha + \frac{1}{\alpha} \geq 2$

hence no such lines are possible.

7. Ans (3.00)

$|x^2 - 2x + y^2 + 1| = |(x-1)^2 + (y-0)^2|$ denotes the square of distance of point (x, y) from $(1, 0)$



minimum distance of $(1, 0)$ from any point

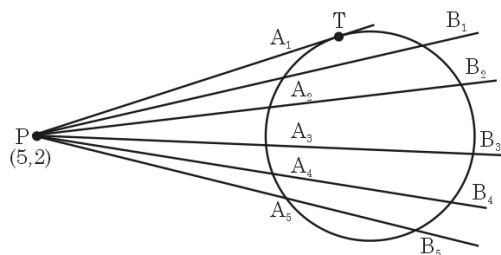
$$(x, y) \text{ on line } 3x + 4y = 6 \text{ is } \left| \frac{3(1) + 4(0) - 6}{\sqrt{3^2 + 4^2}} \right| = \frac{3}{5}$$

$\Rightarrow$ minimum value of the given expression

$$\text{is } 25\left(\frac{3}{5}\right)^2 = 25 \times \frac{9}{25} = 9 = m$$

$$\Rightarrow \frac{m}{3} = 3$$

8. Ans (215.00)



$$P(5, 2) \quad x^2 + y^2 = 25$$

$$PA_1 + PB_1 = 5$$

$$PA_2 + PB_2 = 6$$

$$PA_3 + PB_3 = 7$$

$$PA_4 + PB_4 = 8$$

$$PA_5 + PB_5 = 9$$

$$PA_1PB_1 = PA_2PB_2 = PA_3PB_3 = PA_4PB_4 =$$

$$PA_5PB_5$$

$$= PT^2 = 25 + 4 - 25 = 4$$

$$\sum_{i=1}^5 PA_i^2 + \sum_{i=1}^5 PB_i^2$$

$$= (PA_1^2 + PB_1^2) + (PA_2^2 + PB_2^2) + \dots + (PA_5^2 + PB_5^2)$$

$$= [(PA_1 + PB_1)^2 - 2PA_1PB_1] + [(PA_2 + PB_2)^2 -$$

$$2PA_2PB_2] \dots [(PA_5 + PB_5)^2 - 2PA_5PB_5]$$

$$= [(25 - 8) + (36 - 8) \dots (81 - 8)]$$

$$= (5^2 + 6^2 + \dots + 9^2) - 40$$

$$= 255 - 40$$

$$= 215$$

9. Ans (8.00)

$$S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$$

$$S_1 \equiv x^2 + y = 32$$

$$S_2 \equiv (x - 2)^2 + (y - 2)^2 = 18$$

P, Q lies on intersection of dc of S_1 & S_2

$$\text{DC of } S_1 : S_3 \equiv x^2 + y^2 = 64$$

$$\text{DC of } S_2 : S_4 \equiv x^2 + y^2 - 4x - 4y - 28 = 0$$

Radius of S will be least if PQ is diameter, i.e.

common chord $(x + y = 9)$ of S_3 & S_4 is

diameter

$$\text{Required circle S is : } x^2 + y^2 - 9x - 9y + 17 = 0$$

$$\therefore g = \frac{-9}{2}, f = \frac{-9}{2}, c = 17$$

$$\therefore g + f + c = 8$$

10. Ans (2.00)

$$x = \sqrt{\tan \theta} - 1 \Rightarrow x + 1 = \sqrt{\tan \theta}$$

$$y = \sqrt{\frac{\cos \theta - \sin \theta}{\sin \theta + \cos \theta}} + 2 \Rightarrow$$

$$y - 2 = \sqrt{\frac{\cos \theta - \sin \theta}{\sin \theta + \cos \theta}}$$

$$(2 + 2x + x^2)(5 - 4y + y^2)$$

$$= (1 + (1 + x)^2)(1 + (2 - y)^2)$$

$$= (1 + \tan \theta) \left(1 + \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta} \right)$$

$\Rightarrow$

$$\left(\frac{\cos \theta + \sin \theta}{\cos \theta} \right) \left(\frac{\cos \theta + \sin \theta + \cos \theta - \sin \theta}{\cos \theta + \sin \theta} \right)$$

$$= \frac{(2 \cos \theta)}{(\cos \theta)} = 2$$